RokiCar

Three-wheel omni car build guide



Green hexagonal body, three omni wheels and a removable magnetic lid.

A small car that can move sideways, turn in place and carry a small camera to explore different viewpoints. Three independently driven omni wheels allow forward, backward, sideways and diagonal movement without a steering mechanism.

What you can do

Phone control over Wi-Fi: joystick driving, landscape two-handed control, phone tilt control, speed adjustment, motor calibration, RGB lighting and emergency stop. Basic driving does not require internet access. Reverse trajectory playback and RSSI-assisted return are experimental features, not closed-loop autonomous navigation.

Build sequence

Gather parts, print the enclosure, install motors and wheels, connect the wiring, ask a local AI assistant to upload the three program files, install the phone app, calibrate with the wheels raised, then test slowly on the ground.

01 Parts to buy

For convenience, choose the CyberBrick all-in-one kit and check whether it includes the core board, receiver board, switch, LED adapter, LEDs and cables. To save money, buy the required parts individually. Reuse existing parts if their specifications match.

Part	Qty	Specification / notes
CyberBrick multifunction core board	1	Compatible with the receiver board; runs the project MicroPython firmware. Included in the official kit.
CyberBrick receiver board	1	M1/M2 motor outputs. Do not buy the remote-control transmitter board instead. Included in the official kit.
DRV8833 motor driver	1	Channel A drives the third motor. Select DRV8833 on the product page. About CNY 4.
GA12-N20 geared motor	3	6 V, 200 rpm. Check shaft shape, length and mounting dimensions. About CNY 8 each, CNY 24 total.
47 mm omni wheel	3	Use the recommended wheel that fits the chassis and motor shaft. Linked wheels do not include couplers. About CNY 2.5 each, CNY 7.5 total.
Power switch module	1	Match the receiver power connector and battery plug. Included in the official kit.
LED adapter / splitter	1	CN301-type WS2812 adapter in the project; check pin order. Included in the official kit.
Single RGB LED module	2	Two front lights; requires matching cables. Included in the official kit.
Battery pack	1	Model reference: 14500-cell pack, 7.4 V, 800 mAh. Match actual dimensions and connector. Included in the official kit.
Magnets	10	8 mm diameter, 2 mm thick.
N20 mounting screws	6	M1.6 x 3 mm machine screws; two per motor, confirmed by the author.
Self-tapping screws for larger mounting holes	Prepare 4	BT3 x 6 mm for two 2.4 mm pilot holes. Suggested size; check the clamped-part thickness.
Self-tapping screws for electronics	Prepare 10	BT2.3 x 4 mm for 2.0 mm pilot holes. Use washers to limit insertion in shallow holes.
Wires, connectors and heat-shrink tubing	As needed	According to the assembly.

The driver, three motors and three wheels together cost approximately CNY 35.5. This is not the total car cost: other parts, shipping and printing material are extra. Prices are author-provided estimates; check specifications when ordering.

02 Buying links and fasteners

Purchasing links are provided for reference.

N20 motors: choose 6 V / 200 rpm, buy three
https://e.tb.cn/h.jZcV4WeUt8d2Sem?tk=MxeITKCtN0T

Motor driver: choose DRV8833, buy one
https://e.tb.cn/h.jZc3rSfxJGTOzlx?tk=jM0xTKCHP70

47 mm omni wheels: buy three and matching couplers separately
https://e.tb.cn/h.8ANl6D2rXDkaOyK?tk=3iwCTKBqVp7

Motor screws: six M1.6 x 3 mm screws

Use two machine screws per motor, screwed into its metal threads. Do not use self-tapping screws in the motors or substitute longer screws without checking.

Self-tapping screws for printed parts

Prepare four BT3 x 6 mm screws and ten BT2.3 x 4 mm screws as initial options.

BT3 screw reference
https://asia.store.bambulab.com/products/bt3-socket-head-cap-self-tapping-screws-shcs-new

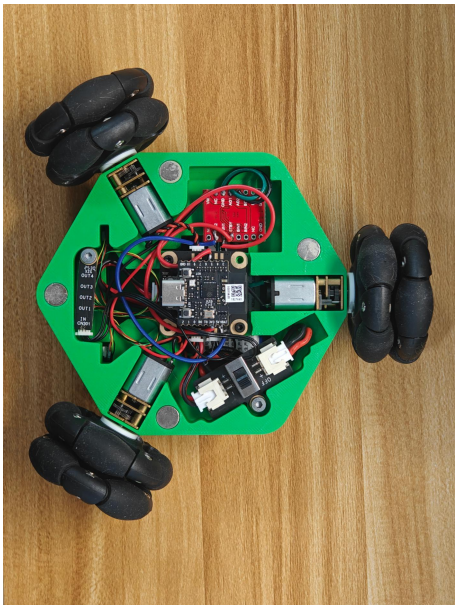
Magnets: 8 mm diameter x 2 mm thick

03 Printing and mechanical assembly

Download the models and print one body shell, one lid and one bottom cover. Use the current model without locating pins and with one magnetic pair for the lid. Do not forget the bottom cover.

Starting settings for a regular FDM printer

Use a 0.4 mm nozzle, 0.2 mm layers and three wall loops. Start with the orientation supplied in the STL; print the lid with its outer top face down. Check bridges, supports and thin-wall paths. Print the magnet-fit test piece first. The designed lid clearance is 0.40 mm per side; actual fit depends on printing accuracy.



Internal layout reference: secure mechanical parts first, then mount boards and organize the wiring.

1. Inspect the prints, magnet pockets and wheel clearance. Remove burrs; do not force magnets into place.

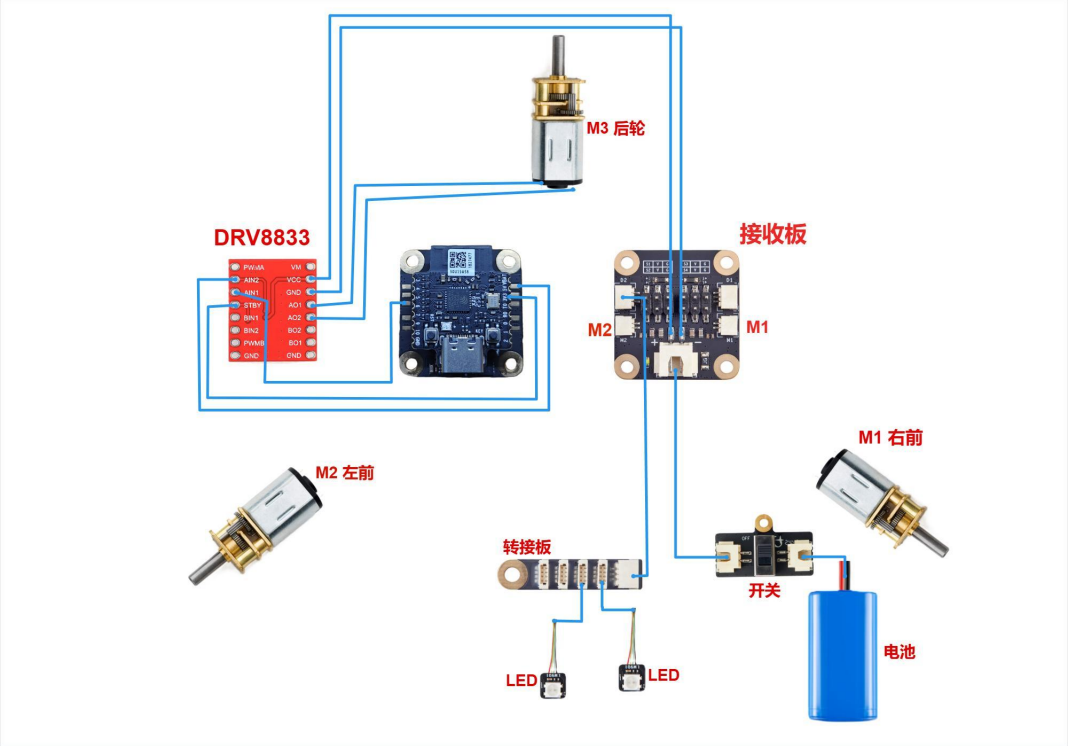
2. Use six M1.6 x 3 mm screws to secure the three N20 motors. Fit the couplers and omni wheels. Check that the wheels are secure, rotate fully and do not rub the shell.

3. Mount the core board, receiver, external motor driver, switch and lights. Choose short self-tapping screws for the holes; use washers to limit screw depth in shallow holes.

4. Place the battery in its compartment and route the wires internally. Do not add an external cable opening, and keep wires clear of the lid.

5. Check the attraction and fix each magnet pair: four pairs for the bottom cover and one pair for the lid. Close both covers and check that the lid can be removed using the finger notch.

04 Wiring



M1: front right; M2: front left; M3: rear. Connect the LED adapter IN to D2. Labels in the original diagram are retained.

Connection	Description
Receiver M1 / M2	Connect to the front-right / front-left motors shown.
DRV8833	Connect to the receiver board power supply.
DRV8833 AO1 / AO2 to M3	Connect the two rear-motor wires; confirm rotation direction through app calibration.
LED adapter IN to D2; OUT to two LEDs	Connect the two front lights through the adapter.
Battery to switch to receiver power	Follow polarity and connector pin order.
Motor power to VM; common GND	Complete VM power wiring for the actual supply arrangement. An omission in the diagram does not mean the connection is unnecessary.
STBY to core board 3.3 V	As specified in the supplied guide.

DRV8833 official chip information and datasheet
https://www.ti.com/product/DRV8833

05 Recommended: ask a local AI assistant to upload firmware

You do not need to type each upload command yourself. Prepare the files, connect the car and give the prompt below to a local AI assistant that can operate your computer.

Choose an AI tool

Examples include Codex, Claude Code, Trae and WorkBuddy. DeepSeek can be used with a local Agent / Harness. The tool must have access to local files, a terminal and the USB serial port. A web-only chat assistant cannot upload files directly to the board.

1. Prepare dependencies

Install your chosen local AI tool and Python 3 on the computer. On Windows, enable the option to add Python to PATH. Uploading uses mpremote; the AI can install it for you.

Python 3 for Windows
https://www.python.org/downloads/windows/

Dependency installation command: py -m pip install --upgrade mpremote

2. Connect the car and provide the firmware folder

Use a USB data cable to connect the CyberBrick core board. Raise all three wheels and close software occupying the serial port. Give the AI the folder containing boot.py, main.py and rokicar_core.py.

3. Copy this prompt to the AI

Please upload the RokiCar three-file firmware in this folder to my CyberBrick core board. Check or install Python and mpremote, identify the actual USB serial port and confirm that MicroPython is already installed. Back up existing programs and calibration parameters, then upload all three files to the device root. Do not change pin definitions or lighting code, and do not erase flash. Read back and compare every uploaded file, then restart. Report whether the upload succeeded and guide me through checking the RokiCar hotspot, phone connection and motor tests with the wheels raised. If an error occurs, diagnose it before continuing.

Confirm the result yourself

The upload is complete only when the RokiCar hotspot appears, the phone app connects, and motor operation and stopping work with the wheels raised. The AI checks file transfer; you must observe the actual movement. Approve serial-port access or commands through the tool interface when prompted.

mpremote documentation for troubleshooting
https://docs.micropython.org/en/latest/reference/mpremote.html

If you do not have an AI tool with local access, use the Thonny method in the last section.

06 Install the app and calibrate before driving

RokiCar project and downloads
https://github.com/yinbaozong/RokiCar

Android APK download
https://github.com/yinbaozong/RokiCar/releases/download/v0.8.0/RokiCar-Controller-v1.22-debug.apk

1. Install the APK on Android and allow installation from that source when prompted. If an old installation uses a different signature, you may need to uninstall it, which clears local settings. There is currently no native iOS app.

2. Connect to Wi-Fi RokiCar, password 12345678. The app connects to ws://192.168.4.1/ws. A no-internet warning is normal.

3. Raise all three wheels and test each port in the app. Map the actual front-right, front-left and rear motors, then correct their directions. Do not infer logical wheel positions from M1/M2 names alone.

4. Calibrate the minimum stable starting PWM and wheel-speed gains, then save. Preview forward motion, sideways motion and rotation; check that the wheels do not rub the shell.

5. Check stopping on joystick release, emergency stop and stopping after connection loss. Then test slowly on the ground. If sideways motion drifts, check mapping, direction and speed calibration rather than increasing speed.



The removable lid can be adapted for a lightweight camera mount or sensor accessories.

Before adding accessories, ensure mounting holes, brackets and ribs cannot contact the battery or wiring. Test balance, load capacity and fastening. The camera is purchased separately; video transmission and gimbal control are not basic features.

A completed build has matching file readbacks, normal power-cycle startup, app connection, correct movement and stopping with the wheels raised, followed by a slow ground test. If no serial port appears, check the data cable and driver. If the port is busy, close other tools. If a module is missing, check filenames and the device root directory.

07 Manual upload: backup method

Extract the matching three-file firmware package first. This uploads Python programs, not a .bin image. The core board must already have compatible MicroPython and provide a USB REPL. Do not erase flash or install a generic ESP32 image just to transfer files.

Thonny: the regular Windows installer includes computer-side Python
https://thonny.org/

Upload all three files

File	Purpose
boot.py	Finite initialization; MicroPython starts the main program after it returns.
main.py	Network communication, motor control, lighting and startup entry.
rokicar_core.py	Motion calculations, command parsing and calibration logic.

1. Connect with a USB data cable, close software using the serial port, and raise the wheels or temporarily disconnect motor power.

2. In Thonny, select the suitable MicroPython (ESP32) interpreter and actual serial port. Do not install or update the underlying MicroPython firmware.

3. Click Stop. If needed, press Ctrl+C in the Shell. At the >>> prompt, enter import sys; print(sys.implementation) to confirm access to MicroPython.

4. Download existing boot.py, main.py, rokicar_core.py and rokicar_config.json to the computer as backups. Record absent files as absent and preserve other factory files.

5. Save the supplied boot.py to the device root, followed by rokicar_core.py and main.py. Keep exact filenames; do not upload only the outer folder or rename files to .py.txt.

6. Download the three files again and compare each against the originals by content or SHA-256. Retransfer mismatches; a successful Save message alone is not sufficient.

7. Close the tool connection, reset or power-cycle, then check the hotspot and app control. If startup loops or the serial port is unstable, restore a stable connection before writing more files.

rokicar_config.json stores motor calibration. Preserve existing calibration; a new car can omit this file, use defaults and save calibration through the app later. It is not the CyberBrick controller-configuration JSON required by the platform.

MicroPython startup documentation
https://docs.micropython.org/en/v1.26.0/reference/reset_boot.html